Analog and Digital Electronics_2023_PYQ

Group-A (Very Short Answer Type Question)

1. Answer any ten of the following:

(1) In the astable amplifier, up to what voltage the capacitor charges?

(2) The state 1110 is a valid state in 8-4-2-1 Binary Coded Decimal counter.
State True / False.

(3) A 4-bit mod 16 ripple counter uses JK flip-flop. If the propagation delay of each flip flop is 100 ns, what will be the maximum clock frequency (in MHz) that can be used in the counter?

(4) In Asynchronous circuit Race condition always arises. State True / False.

(5) What is the minimum number of NAND gates required to design a Full adder circuit?

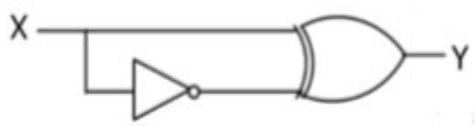
(6) Which type of ADC (analog-to-digital converter) is slowest of all?

(7) Which A/D converters are used for High speed operation?

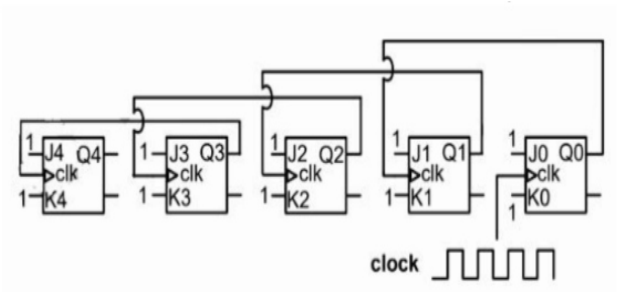
(8) The input of a Schmitt Trigger is sawtooth wave, what will be the shape of the output?

(9) The output Y of the logic circuit given below is __.

versity.onrender.com



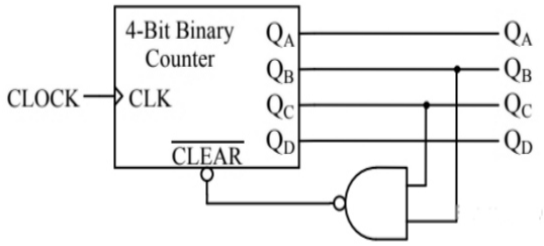
(10) Five JK flip-flops are cascaded to form circuit shown in the figure. Clock pulses at a frequency of 1 MHz are applied as shown. What will be the frequency (in kHz) of the waveform at Q3?



(11) If $(212)_x$ - $(23)_{10}$, where X is the base of the number system, find X.

(12) A mod-n counter using a synchronous binary up-counter with synchronous clear input is shown in

the figure. What will be the value of n?



Group-B (Short Answer Type Question)

Answer any three of the following:

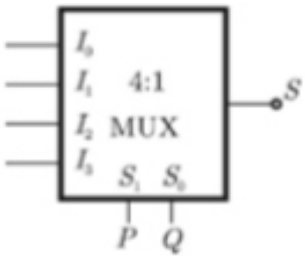
2. Perform Conversion of D flip Flop to J-K Flip Flop.

3. Perform conversion of S-R Flip flop to J-K Flip Flop.

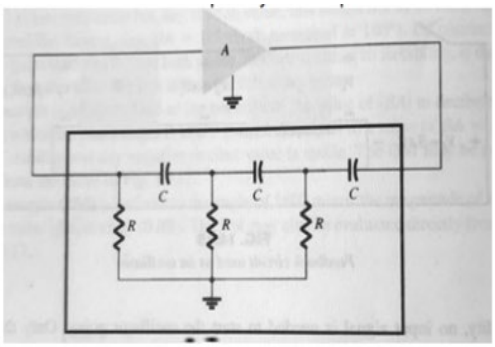
4. Design a 5 to 32 decoder using 3 to 8 decoder and 2 to 4 decoders.

5. Figure shows a 4 to 1 MUX to be used to implement the sum S of a 1-bit full adder with input bits P and Q and the carry input Cin. Find the combinations of inputs to I_0, I_1, I_2 and I_3 of the MUX which will realize the sum S.

versity.onrender.com



6. Find the oscillation frequency f of the phase shift oscillator when R = 10KΩ and C = 6.5nF.

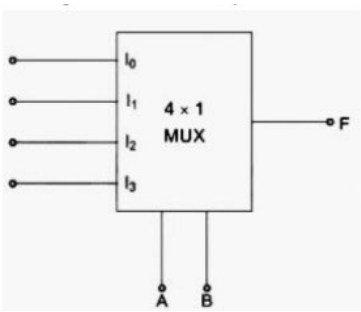


Group-C (Long Answer Type Question)

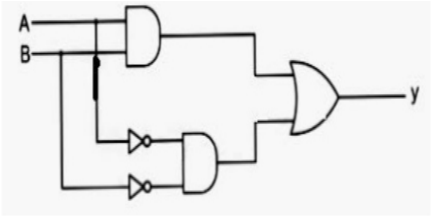
Answer any three of the following:

7.

(a) In the given MUX the output is F = A XOR B. What will be the inputs I_0, I_1, I_2, I_3?

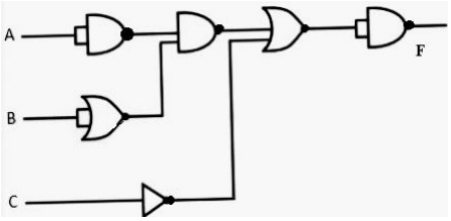


(b) Check the following digital circuit. What will be the equivalent logic gate of the whole circuit?



vers er.com

(c) What will be minimized output F of the circuit?



8.

(a) Define Voltage, Current, and Power amplifier.

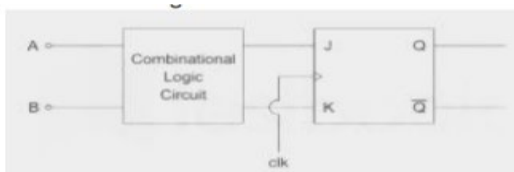
(b) Write the different classes of Power amplifier.

(c) Draw the circuit diagram of a class B push pull transistor amplifier. Explain the operation.

(d) Derive the maximum efficiency of a class B amplifier.

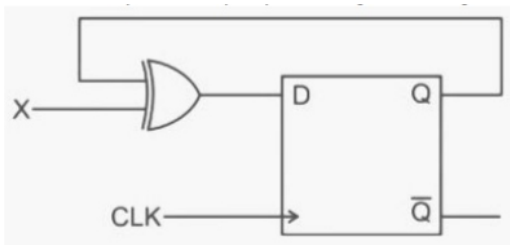
9.

(a) To realize the given truth table from the circuit shown in the figure, find the input to J in terms of A & B



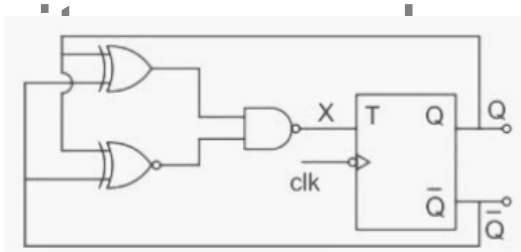
A	B	Q_{n+1}
0	0	$\overline{Q_n}$
0	1	1
1	0	Q_n
1	1	0

(b) What is the equivalent flip-flop of the digital circuit given below; explain with reason.



(c) The clock frequency applied to the digital circuit shown in the figure below is 1KHz. If the initial state of the output of the flip-flop is 0, what will be the frequency of the output waveform Q in KHz?

ver om

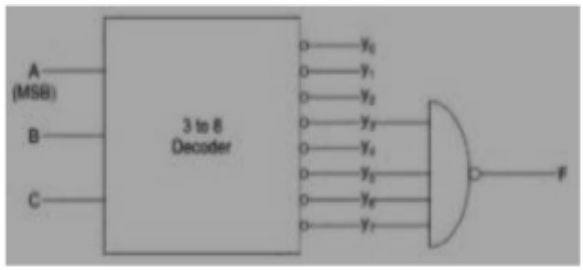


10.

(a) Find the simplified form of the Boolean expression Y, where

Y=(A'.BC+D)(A'.D+B'.C')

(b) The output F of decoder can be realized with best logic gates. Find out the Boolean expression.



11.

(a) Draw the circuit diagram of an Astable Multivibrator using IC 555 timer.

(b) Derive the expression duty cycle and frequency of oscillation.

(c) Explain why this is called Free running oscillator.

versity.onrender.com